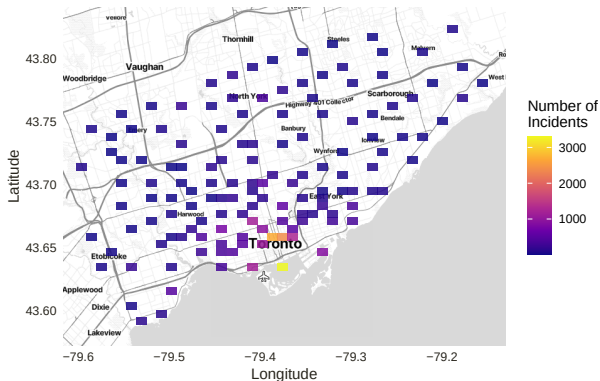


Bicycle Theft Spatial Risk Prediction

A Multi-Model Comparison: Baseline, Linear, IDW and Kriging

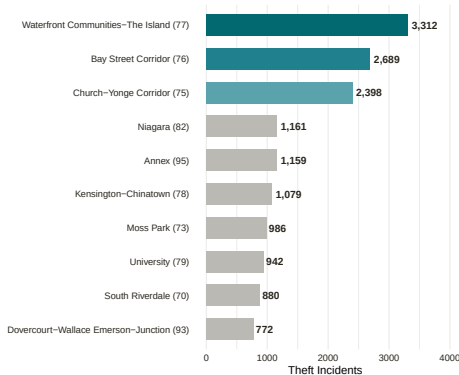
Spatial Distribution of Bike Theft Incidents

Each grid cell represents the count of thefts in that area



Top 10 High-Risk Neighborhoods

Top 10 Neighborhood Theft Counts (2014-2023)

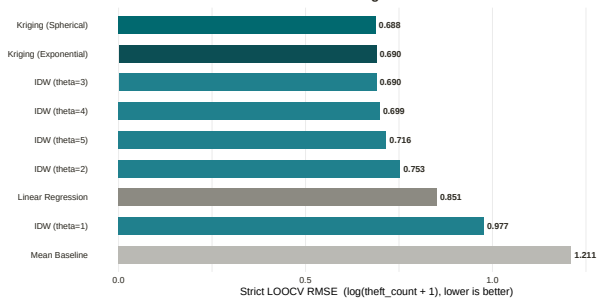


31,833 incidents; **140** neighborhoods; top 3 = **25.9%**; Moran's $I = 0.288$ ($p < 0.001$).

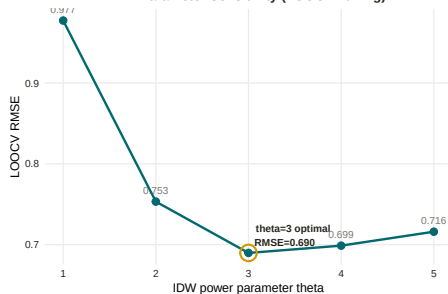
Which Spatial Model Best Explains Theft Risk?

Before choosing a method, we ask: **is this dataset suited to variogram-based kriging?** All 31,833 records collapse to **140 unique coordinates** — neighborhood *centroids*. We tested **9 candidate models** with strict spatial LOOCV on $\log(\text{theft_count} + 1)$; kriging variograms are refit inside each held-out fold.

LOOCV Model Ranking – 9 Candidate Models



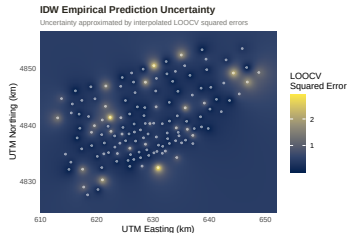
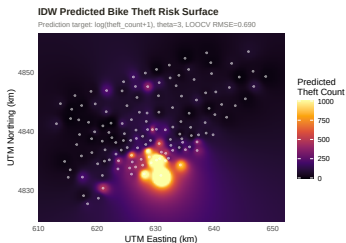
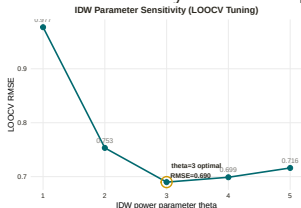
IDW Parameter Sensitivity (LOOCV Tuning)



Kriging (Spherical) and **IDW ($\theta = 3$)** are statistically indistinguishable: 0.688 vs. 0.690 RMSE, a 0.3% gap. Both cut baseline error by **43%**. **IDW is chosen for robustness, not because its tuned CV score is uniquely best.**

Model Selection: IDW Chosen for Robustness

- Kriging (Spherical) wins strict LOOCV RMSE by a razor-thin margin.
- **IDW is selected**: no distributional assumptions, no fold-to-fold variogram refits, one interpretable parameter (θ), and stable across tuning.
- Uncertainty uses interpolated LOOCV squared errors because IDW has no native variance estimator.

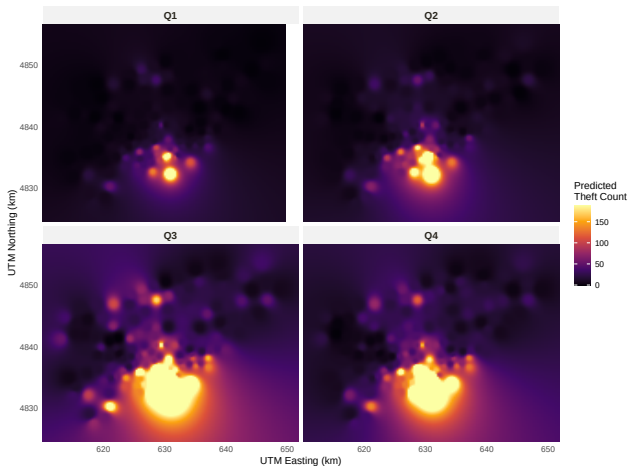


IDW tuning RMSE stays within $\pm 4\%$ of optimal across $\theta = 2-5$: the model is **not fragile to tuning**.

Confidence is lowest at sparse boundaries, where new sampling would help most.

[Interactive Heatmap](#) | [LOOCV Dashboard](#) | [Full Code](#) | [Data](#) | [Full Report](#)

Seasonal Hotspots and Recommendations



[Interactive Heatmap](#) | [LOOCV Dashboard](#) | [Full Code](#) | [Data](#) | [Full Report](#)

Key findings

- **Q3 (41.9%) + Q4 (33.3%) = 75%** of incidents; Q1 is 9%.
- The waterfront hotspot persists and **intensifies in Q3–Q4**.
- IDW and kriging agree on the same high-risk core.

Recommendations

1. Patrol + secure parking downtown before Q3.
2. Sample sparse boundary areas.
3. Use IDW ($\theta = 3$).

Supplementary Materials

Interactive Heatmap:

[GitHub Pages / index_mywebsite](#)